

SIMPLE IMPACT STUDIES — SAMPLE DRAFT

Harris County Sample

Traffic Impact Study

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Simple Impact Studies — sample draft

Traffic Impact Study

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EXECUTIVE SUMMARY

This Traffic Impact Analysis (TIA) presents the anticipated traffic impacts of the proposed Harris County Sample located within Houston-The Woodlands-Sugar Land MSA, Texas. The study evaluates 14 intersections within a 1.00-mile study radius. The report follows the outline in TxDOT Traffic and Safety Analysis Procedures Manual (TSP) Chapter 16 Appendix Q, with capacity analysis by the openly-published Webster/Akcelik signalized control-delay method that the TSP capacity procedures are built on, and trip generation per the public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716). The development generates a determining peak-hour trip count of 680 vehicles, classifying it as Category 2 (500–1,000 peak-hour trips) under TSP §16.2.1 Table 16-1.

Reviewing authority: City of Houston.

Houston Public Works — Office of the City Engineer (OCE), Traffic Group, per the Infrastructure Design Manual (IDM) Ch. 15, revision 07-01-2022, and the OCE TIA Content Guide.

Findings:

- 2 intersections are projected to drop one or more LOS grades under the Background-plus-site scenario.
- 3 intersections operate at LOS E or F under the Background-plus-site scenario and are flagged for mitigation under TSP §16.4.3.

14 INTERSECTIONS	2 LOS DROPS	3 AT LOS E/F	55.1s WORST Δ DELAY
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1.0 INTRODUCTION

Per TSP §16.1, this Traffic Impact Analysis determines whether the transportation infrastructure surrounding the project can accommodate the traffic demand the proposed development will introduce. It is layered with the City of Houston TIA standard where applicable. Houston Public Works — Office of the City Engineer (OCE), Traffic Group, per the Infrastructure Design Manual (IDM) Ch. 15, revision 07-01-2022, and the OCE TIA Content Guide.

TSP §16.1 also flags that this chapter does not provide final TIA recommendation thresholds — the TxDOT District and the host municipality may request analysis beyond this scope, and the project manager determines final methodology. Refer to the TxDOT Access Management Manual Chapter 3 for the TIA-request criteria that apply on the state system.

Preliminary Scoping (TSP §16.2.2)

Per TSP §16.2.2, the specific project scope should be confirmed with the local TxDOT District before the TIA begins, via an in-person meeting or alternate correspondence. Items to confirm:

- Number of access points to TxDOT roadways
- Acceptable LOS thresholds
- Selected study years
- Anticipated influence area
- Intersections and roadways to be analyzed
- Scenarios to analyze
- Data collection method
- Project schedule and buildout year
- Data source
- Use of TDM outputs, growth factors, etc.
- Other major projects in the area

Required submission: TIA + Access Management Data Summary Form (mandatory for commercial sites, embedded in IDM pp. 15-5 to 15-8) + CPC 101 Form (per OCE TIA Content Guide p. 3),

submitted via the Houston Permitting Center. Approval is required before plan submittal if no plat is required. The TIA must be sealed by a Texas-licensed civil PE (civil specialty per IDM §15.04.B.1.a, not traffic specialty).

Host-jurisdiction TIA threshold: ≥ 100 PHT triggers a scoping meeting that determines whether a full TIA is required (IDM §15.04.A.4.a, revision 07-01-2022). The Technical Memorandum tier is 80–120 vph during AM or PM peak (IDM §15.04.A.5). Table 15.04.01 categories: I (<100 PHT), II (100–499), III (500–999), IV ($\geq 1,000$).

2.0 PROJECT DESCRIPTION

2.1 Site Plan

Project name	Harris County Sample
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	30.0080°, -95.5210°
Host jurisdiction	City of Houston

2.2 Area of Influence

The anticipated area of influence is the 1.00-mile radius surrounding the site, defined per TSP §16.3.1 to include the roads and intersections within the project and the area impacted by project-generated trips. Final area of influence is confirmed with the District during preliminary scoping.

2.3 Phasing and Timing

Single-phase buildout year: 2027. Future analysis horizon for this TIA category: Buildout year, each phase completion year, and five years past buildout (TSP §16.2.1 Category 2).

Phased development scenarios are not modeled in this screening analysis. Each phase completion year listed above should be analyzed separately in the formal submittal — phase boundaries and completion dates must come from the applicant's construction schedule.

3.0 STUDY AREA CONDITIONS

3.1 Existing and Anticipated Land Use

Existing land use and the host-jurisdiction Future Land Use Plan for parcels within the area of influence should be compiled from the host city's adopted Comprehensive Plan and any applicable overlay districts. This screening report does not generate that inventory [placeholder — requires GIS pull against host-jurisdiction parcel layer].

3.2 Existing and Future Roadway System

Roadway functional class, lane count, posted speed, and surface for state-system routes (IH / US / SH / FM / RM / BU / BS / SL / SS) are referenced against TxDOT Roadway Inventory (RHiNo) and the TxDOT Statewide Planning Map. Future roadway system context is taken from the TxDOT Unified Transportation Program (UTP) and the regional MPO Metropolitan Transportation Plan (MTP) where in-area projects are programmed.

4.0 EXISTING OPERATIONS

4.1 Roadway Conditions and Traffic Controls

Geometry, signal timing, and traffic-control inventory for the study network are derived from RHiNo plus host-jurisdiction signal-timing records. For the formal submittal, supplementary field inspection within 12 months of submittal is recommended.

4.2 Alternate Modes

Pedestrian, bicycle, and transit facility inventories within the area of influence should be confirmed against host-jurisdiction GIS and regional transit-operator maps. Per TSP §16.3.3, multimodal reduction is applied to trip generation in areas where alternate transit is readily available — this screening report does not auto-apply multimodal reduction.

4.3 Traffic Volumes

Existing AADT for state-system segments is taken from the TxDOT Open Data Portal AADT layer (annual refresh). Peak-hour turning-movement counts at study intersections should be collected mid-

week (Tue/Wed/Thu), school-in-session, within 12 months of submittal — per Houston IDM §15.06.01.A counts must be within 12 months in high-growth areas and within 24 months elsewhere; Austin TDS no longer accepts pre-COVID counts by default.

4.4 Level of Service

Affected intersection	Distance (mi)	Existing LOS	Existing delay (s)
Near Cypresswood Drive	0.02	C	32.4
Cypresswood Drive & Squyres Road	0.25	A	6.6
Cypresswood Drive & Stuebner Airline Road	0.40	B	15.7
Stuebner Airline Road & Wimbleton Estates Drive	0.58	A	7.9
Lyons School Road & Stuebner Airline Road	0.60	A	6.6
Cypresswood Drive & York Minster Drive	0.65	A	6.6
Stuebner Airline Road & Strack Road	0.75	F	202.8
Cypresswood Drive & Champion Forest Drive	0.84	A	6.7
Louetta Road & Kleinwood Drive	0.85	A	6.6
Louetta Road & Lyons School Road	0.87	A	6.6
Squyres Road & Louetta Road	0.87	A	6.8
Louetta Road & Theiss Mail Route	0.89	E	77.7
Herts Road & Champion Forest Drive	0.90	B	11.6
Louetta Road & Stuebner Airline Road	0.91	C	23.5

4.5 Safety

Crash history for the most recent five years on study segments and intersections should be pulled from the TxDOT Crash Records Information System (CRIS Public Query). This screening report does not auto-generate the crash summary [placeholder — requires CRIS pull].

5.0 PROJECTED TRAFFIC

5.1 Site Generated Traffic

Trip generation is calculated per the public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716) for land use 820 (Shopping Center / Retail) at a proposed size of 85 1,000 sqft GLA. Per TSP §16.3.3, the fitted curve equation is preferred when the data plot contains at least 20 data points or has an R^2 of at least 0.75; otherwise average rates apply. Daily, AM peak, and PM peak hour trips are reported below.

Period	Entering trips	Exiting trips
Daily	3,400	3,400
AM peak hour	166	106
PM peak hour	326	354

Internal capture (per TSP §16.3.3: trips between land uses within the same development that do not touch the off-site street system) and pass-by trips (already traveling on the adjacent roadway network and entering the proposed development) are accounted for below.

Pass-by capture applied 30%

Internal capture applied

0%

Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

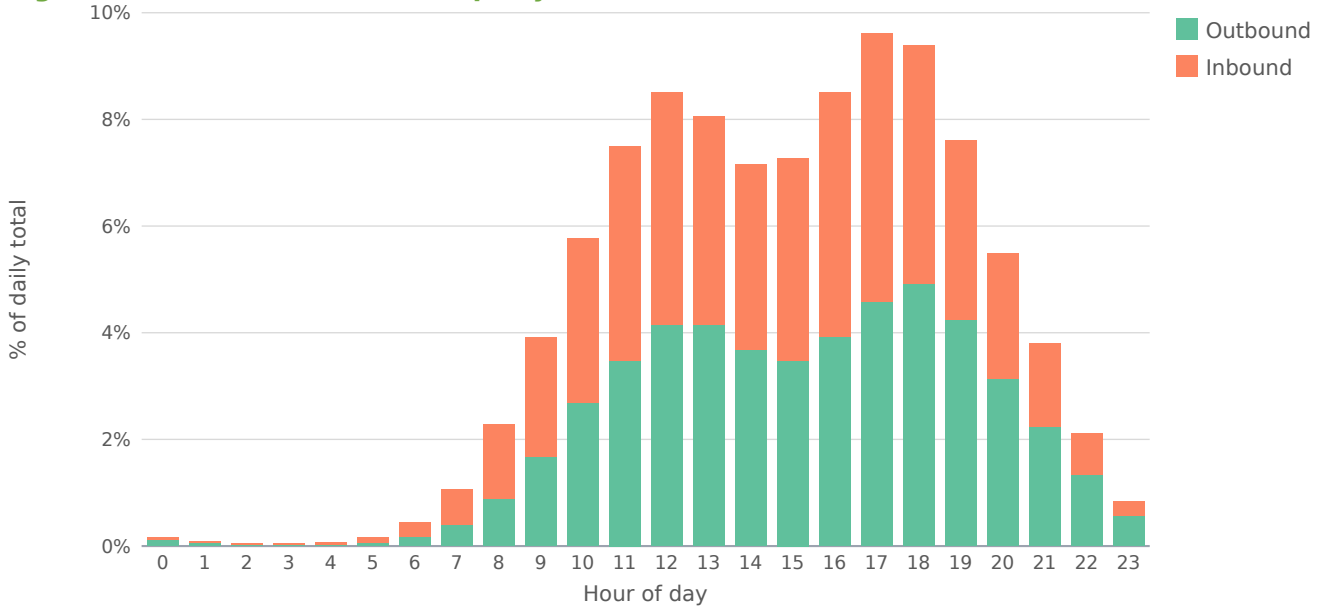
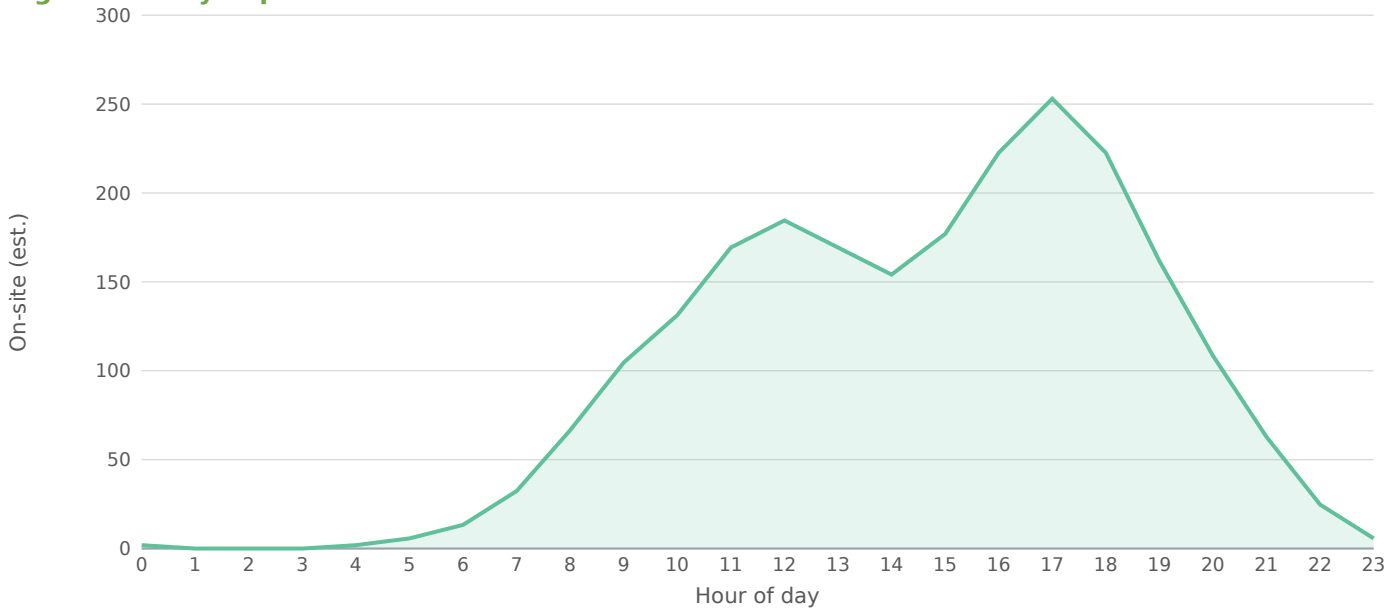


Figure — Daily Trip Accumulation



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

5.2 Trip Distribution

Per TSP §16.4, project distribution is assigned using engineering judgment, informed by the surrounding roadway network geometry and proximity to the project access points. If only one project driveway is proposed, all trips enter and exit through that driveway. Final distribution percentages are confirmed with the District during preliminary scoping.

5.3 Trip Assignment

Site-generated external trips are assigned to the study network using inverse-distance weighting from the project site to each signalized intersection, normalized so the period total matches the external-trip count from §5.1.

5.4 Non-Site Traffic

Background traffic is grown at 1.50% per year, applied as an explicit override, not derived from measured data. Explicit override — 1.50%/yr was supplied with the request and applied in place of the engine's own rate. No measured growth rate is wired for this region, so there is no measured value to compare the override against. The applied rate is not derived from measured data; its basis must be stated by the preparer. Per TSP §16.4.2, the prescribed method is to average at least the last five years of historical AADT data for the segment analyzed; the rate applied here should be refined to per-segment trend on the affected facilities before formal submittal. Background growth data is also commonly sourced from the host city or regional MPO travel-demand model (H-GAC, NCTCOG, CAMPO, or AAMPO depending on the MSA).

Other major projects in the area (committed developments) must be coordinated with the District and governing municipality per TSP §16.3.3 and added on top of the AADT-derived background growth. This screening analysis does not auto-pull committed-development trips.

5.5 Total Traffic

Period	Raw	Pass-by	Int. cap.	External	In	Out
AM Peak Hour	272	20	0	211	129	82
PM Peak Hour	680	204	0	399	191	208
Saturday Middyay	748	56	0	580	290	290
Daily Total	6,800	510	0	5,269	2,635	2,634

The external / net-external column is net-new vehicle trips assigned to the roadway = (gross – pass-by – internal-capture) × auto-mode share (83.7% for this metro). Walk / bike / transit person-trips are excluded from off-site assignment, so In + Out equals the external column.

5.6 Trip Distribution — Gravity Model

The trip distribution uses the gravity model (term = $M / (d^1 \cdot d_{\text{site}})$; mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector	Share of project trips
Northeast (NNE + ENE)	26%
Southeast (ESE + SSE)	12%
Southwest (SSW + WSW)	22%
Northwest (WNW + NNW)	40%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Louetta Road & Theiss Mail Route	WNW	0.89	3,164	0.2	17.14%
Stuebner Airline Road & Wimbledon Estates Drive	NNE	0.58	2,197	0.1	13.24%
Near Cypresswood Drive	SSW	0.02	1,732	0.1	12.76%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Stuebner Airline Road & Strack Road	ESE	0.75	2,132	0.1	12.18%
Louetta Road & Stuebner Airline Road	NNW	0.91	2,197	0.1	11.90%
Cypresswood Drive & Stuebner Airline Road	ENE	0.40	1,732	0.1	11.10%
Cypresswood Drive & Champion Forest Drive	WSW	0.84	1,579	0.1	8.76%
Herts Road & Champion Forest Drive	WNW	0.90	1,579	0.1	8.59%
Squyres Road & Louetta Road	WNW	0.87	250	0.0	1.38%
Cypresswood Drive & Squyres Road	WSW	0.25	100	0.0	0.68%
Lyons School Road & Stuebner Airline Road	NNE	0.60	100	0.0	0.60%
Cypresswood Drive & York Minster Drive	ENE	0.65	100	0.0	0.59%
Louetta Road & Kleinwood Drive	NNW	0.85	100	0.0	0.55%
Louetta Road & Lyons School Road	NNW	0.87	100	0.0	0.55%

14 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

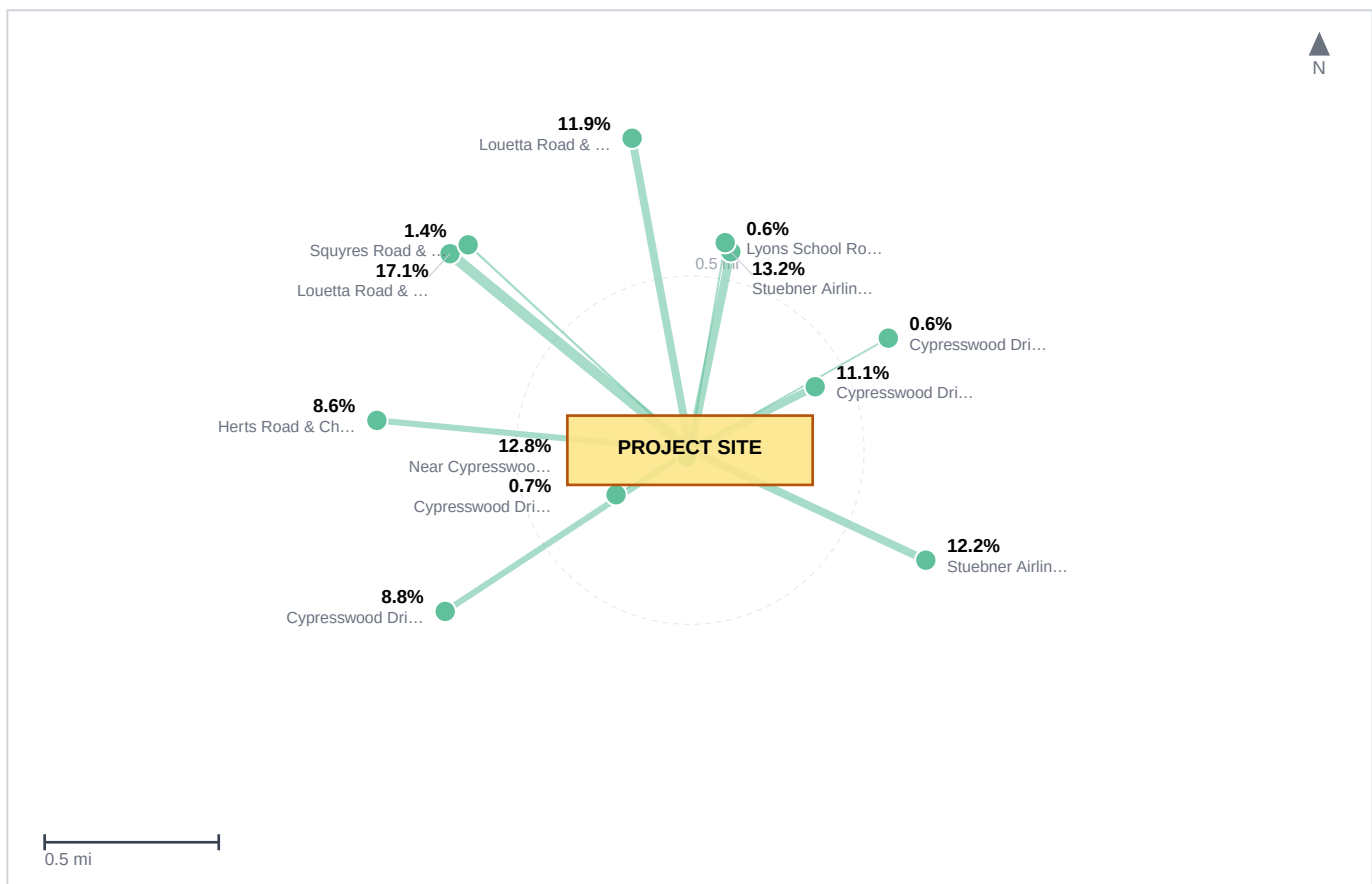
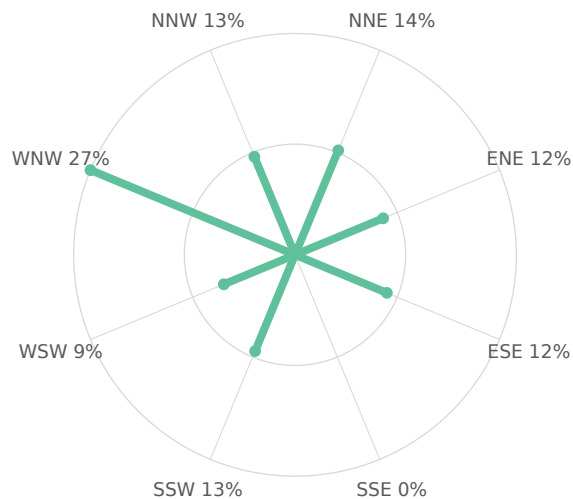


Figure — Project Trip Distribution. Study-area zones plotted to scale at their true bearing and distance from the site; leg weight is proportional to each zone's share of project trips, and the label gives that share. Derived from the Gravity Model distribution — the same shares tabulated above. Screening-grade: zone positions are the analysis locations, not a surveyed base map.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length $\propto$ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

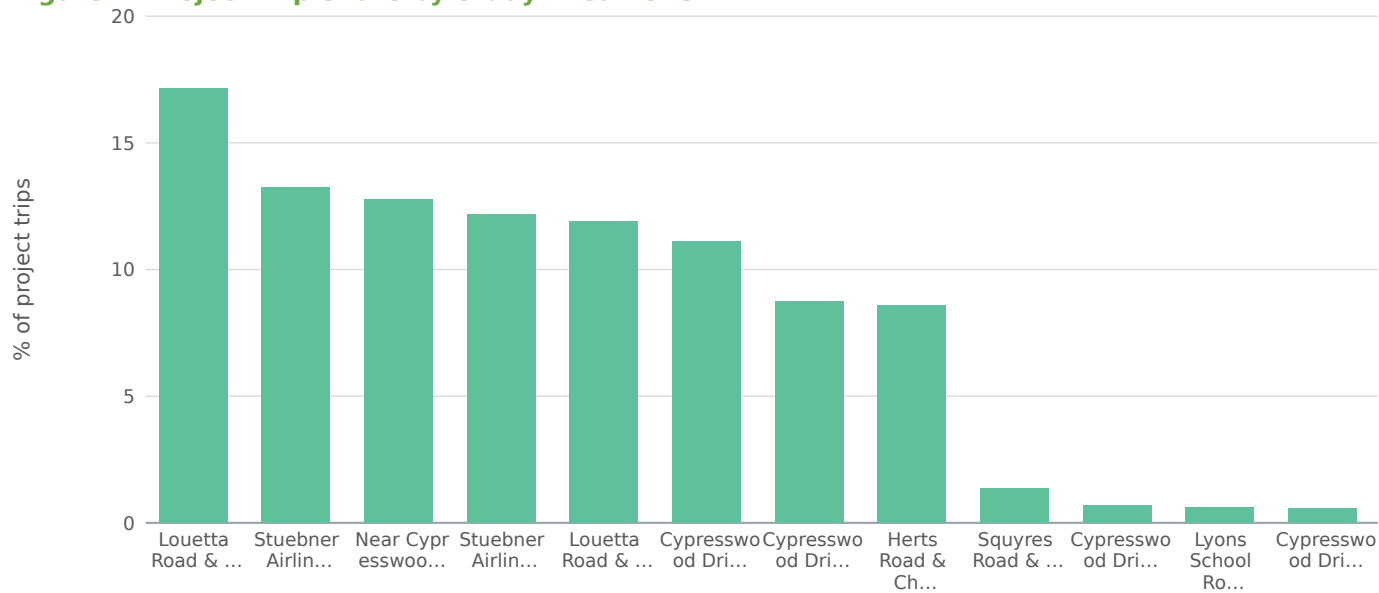


Figure — Trip Share vs. Distance from Site (Gravity Decay)



6.0 TRAFFIC AND IMPROVEMENT ANALYSIS

6.1 Site Access

Where the project fronts a state-system roadway (IH / US / SH / FM / RM / BU / BS / SL / SS), site access is reviewed through the TxDOT Driveway Access Permit (DAP) process under the Access Management Manual Chapter 3 §3. Driveway spacing, geometry, and auxiliary lanes (right-turn deceleration, left-turn) are checked against ACM Chapter 2 §3 (Table 2-2 spacing) and the Roadway Design Manual Chapter 16.

6.2 Auxiliary Lane and Sight Distance Analysis

Auxiliary lane warrants (right-turn deceleration, left-turn) per ACM Chapter 2 §4 and Roadway Design Manual Chapter 16 should be checked against the project's access geometry. Intersection sight distance per RDW Ch. 2 should be verified at every proposed driveway. This screening report does not perform per-driveway sight-distance or auxiliary-lane warrant calculations [placeholder — requires final driveway geometry].

6.3 Capacity and Level of Service Analysis

Per TSP §16.4.1 and §16.4.4, submittal capacity analysis follows the latest HCM methodology using one of the District-accepted tools (Synchro, HCS, Vissim, or Vistro); this screening applies the openly-published Webster/Akçelik signalized model pending that tool run. For signalized intersections, each approach and the overall intersection are analyzed. Two future scenarios are compared at each affected intersection: Background (grown traffic without the proposed project) and Background-plus-site (grown traffic plus the project's external trips at the assigned distribution). An Existing scenario is included for context. Host-jurisdiction LOS standard: Per IDM §15.04.B.6.a: the need for mitigation is determined by using the qualitative measure Level-of-Service (LOS); the threshold of significance for transportation facilities on the area street system is LOS D.

Intersection	Existing LOS	Background LOS	Bgd+Site LOS	Δ delay (s)	Q95 (ft)
Near Cypresswood Drive	C	C	▲ F	55.1	393
Cypresswood Drive & Squyres Road	A	A	A	0.5	169
Cypresswood Drive & Stuebner Airline Road	B	B	B	2.3	973

Intersection	Existing LOS	Background LOS	Bgd+Site LOS	Δ delay (s)	Q95 (ft)
Stuebner Airline Road & Wimbledon Estates Drive	A	A	A	0.5	825
Lyons School Road & Stuebner Airline Road	A	A	A	0.4	121
Cypresswood Drive & York Minster Drive	A	A	A	0.1	121
Stuebner Airline Road & Strack Road	F	F	F	11.9	1,133
Cypresswood Drive & Champion Forest Drive	A	A	A	0.1	395
Louetta Road & Kleinwood Drive	A	A	A	0.2	121
Louetta Road & Lyons School Road	A	A	A	0.2	121
Squyres Road & Louetta Road	A	A	A	0.4	146
Louetta Road & Theiss Mail Route	E	F	F	10.6	1,696
Herts Road & Champion Forest Drive	B	B	B	0.1	420
Louetta Road & Stuebner Airline Road	C	C	▲ D	12.0	825

6.4 Mitigation

Per TSP §16.4.3, any operational deficiencies found in the future analysis are considered for mitigation. The threshold for acceptable operations (LOS, queue length, travel time, and other MOEs) is agreed upon with the District during preliminary scoping rather than being set as a single statewide standard. Typical mitigation measures listed in TSP §16.4.3 include: right-turn deceleration lanes, left-turn lanes, median modifications, traffic signal modification and installation, road widening, revised striping, turning lane restrictions, and alternative intersections / interchanges. The developer is responsible for implementing the agreed-upon mitigation measures.

Screening Mitigation Recommendations

Near Cypresswood Drive [MAJOR]

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +55.1s, design year +83.6s).

Cypresswood Drive & Stuebner Airline Road [MINOR]

Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required. Governing horizon: design year (opening year +2.3s, design year +7.6s).

Stuebner Airline Road & Strack Road [MAJOR]

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +11.8s, design year +0.0s).

Louetta Road & Theiss Mail Route [MAJOR]

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +10.6s, design year +12.7s).

Louetta Road & Stuebner Airline Road [MAJOR]

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +12.0s, design year +28.5s).

Houston commercial submittals must include the Access Management Data Summary Form (IDM pp. 15-5 to 15-8) and the CPC 101 Form (OCE TIA Content Guide p. 3) alongside this TIA; an MDR drainage report integration is required where new impervious cover is added — none of these are

generated by this screening report [placeholder — requires site-civil inputs].

6.5 Driveway Operational Analysis

Per-driveway operational analysis (full-movement vs. left-in/left-out configuration, throat depth, on-site queue spillback) depends on the final site plan and is not included in this screening-level TIA [placeholder].

7.0 SAFETY ANALYSIS

Per the Appendix Q outline, Safety Analysis is a standalone section separate from §4.5 Existing Operations — Safety. It includes the project's effect on study-area crash trends, sight-distance impacts at proposed access, and any HSIP-identified crash clusters within the area of influence. This screening report does not auto-generate the safety analysis [placeholder — requires CRIS pull + sight-distance verification].

8.0 CONCLUSIONS

Based on the screening analysis above, the project is classified as Category 2 (500–1,000 peak-hour trips) per TSP §16.2.1. Of the 14 affected intersections analyzed, 2 drop one or more LOS grades and 3 operate at LOS E or F under the Background-plus-site scenario. The horizon analyzed in this screening is the buildout year only; the full submittal must cover the years required by the project's TIA category per TSP §16.2.1 Table 16-1.

9.0 RECOMMENDATIONS

Submit this TIA to City of Houston, with parallel TxDOT-District coordination where any state-system route is in frontage. Validate the screening results against current-edition manuals, updated turning-movement counts within the most recent 12 months (Houston IDM §15.06.01.A: 12 months in high-growth areas, 24 months elsewhere), and the preliminary-scoping outcome with the District. The report must be sealed by a Texas-licensed Professional Engineer per 22 TAC §137.33 (Sealing Procedures), promulgated under The Texas Engineering Practice Act (Tex. Occ. Code Ch. 1001), with the seal on the cover and on every sealed sheet — Houston IDM §15.04.B.1.a requires the signing PE to hold the civil specialty (not traffic).

10.0 APPENDICES

The formal submittal appendices, per the Appendix Q outline, should include: scoping correspondence with the TxDOT District; site plan figures; TMC count sheets with date, weather, observer; Trip-generation worksheets with rate/equation selection rationale; HCS / Synchro / Vissim / Vistro output; signal warrant analyses citing the TMUTCD (2025 edition, effective Jan 18, 2026); auxiliary-lane and sight-distance worksheets; CRIS crash data summary; and the host-jurisdiction's submission forms (Houston Access Management Data Summary Form + CPC 101 Form / Austin TIA Determination + Scope / Dallas TIS Waiver / Fort Worth TIA Worksheet / San Antonio TIA Threshold Worksheet + Rough Proportionality calculation).

Programmed Projects (informational)

Review of programmed transportation projects within the area of influence should consult the TxDOT Unified Transportation Program (UTP 2026, adopted Aug 2025), the federally-required Statewide Transportation Improvement Program (STIP), and Houston-Galveston Area Council (H-GAC) TIP 2025–2028. This screening analysis does not automatically integrate programmed-projects data; manual review is recommended for any submittal.

FINDINGS

- Project will generate 6,800 new daily vehicle trips, with 680 during the PM peak hour (326 inbound / 354 outbound).
- Pass-by credit 30% and internal-capture credit 0% applied at the PM peak (25% of that credit at the AM and Saturday-midday periods, per the industry rule of thumb that off-peak shopping diverts less) before off-site assignment (standard pass-by methodology; ULI Internal Capture).
- Auto-mode share of 84% applied to external trips for Houston-The Woodlands-Sugar Land MSA; 16% of generated trips are assumed to arrive by transit, walking, or cycling and do not load the off-site roadway network. Source: ACS 5-Year Estimates Table B08301 (US) or national-stats equivalent.

- Existing volumes were grown by 1.50%/yr over 1 year ($\times 1.01$) to the opening-year horizon.
- 14 signalized intersections fall within the study area; 2 are projected to drop at least one LOS grade in the opening year after build-out.
- 3 intersections are projected to operate at LOS E or F under the opening-year build condition and require formal mitigation per Houston Public Works (Traffic Operations) TIS guidance.
- Worst opening-year impact: Near Cypresswood Drive — projected delay rises 55.1s (LOS C $\rightarrow$ F); 95th-pct queue 393 ft on the critical approach.
- Largest projected delay change: 55.1s in the opening year (Near Cypresswood Drive) and 83.5s in the design year (Near Cypresswood Drive). The design-year figure is the larger; that difference is driven by background traffic growth over the design horizon, not by the development.
- Conclusions are reported at the applied screening assumptions. Discrete-scenario sensitivity at the formal TIA scoping meeting should test: trip-generation method (rate vs. fitted-curve equation); internal capture and pass-by credit variants; and a $\pm 0.5\%$ /yr background growth band around the applied value.

METHODOLOGY NOTES

- Trip generation uses public-data average rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) for the selected land-use code, computed for AM peak, PM peak, Saturday midday, and daily totals. Saturday-midday rates are estimated as a published industry multiple of the PM peak rate by land-use category.
- Pass-by and internal-capture credits are applied in full at the PM peak (and at 25% of that credit fraction for the AM and Saturday-midday periods) per standard pass-by screening methodology and ULI Mixed-Use Internal Capture defaults; only the residual external vehicle trips are assigned to off-site intersections.
- Existing intersection volumes are grown to the opening-year horizon at the user-supplied annual growth rate (default 1.5%/yr) before the capacity analysis.
- Weather adjustment applies published rain/snow capacity-reduction factors as adopted in US traffic-engineering practice: clear 1.00, light rain 0.95, heavy rain 0.86, light snow 0.86, heavy snow 0.70. The factor multiplies the saturation flow at every intersection.
- Off-site impact is screened for all signalized intersections within the study radius (default 0.5 mi) using the four-step travel demand model (FHWA; NCHRP Report 716). Step 1 Trip Generation: public-data average rates (SANDAG 2002 / NHTS 2017 / NCHRP 716) give the site's external (post pass-by / internal-capture) productions. Step 2 Trip Distribution: a production-constrained gravity model $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$ allocates trips to surrounding signals, where attractiveness A_j is the signal's through-volume and the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work coefficients $a=28507$, $b=-0.02$, $c=-0.123$) on the travel time t to each signal. Step 3 Mode Choice: a binary logit $P(\text{auto}) = 1 / (1 + e^{-(\text{ASC} - \lambda \cdot \Delta \text{GC})})$ calibrated to the metro's auto-mode share — measured (ACS B08301) where the metro is wired, otherwise the ACS B08301 metropolitan median of 90% — and shifted by site urbanity (a density proxy from surrounding through-volumes) so denser, more transit-served sites split further from auto; only the resulting vehicle trips load the network. Step 4 Route Assignment: a capacity-constrained assignment using the BPR volume-delay function $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives. Signals lacking AADT data fall back to a constant 5,000 vpd attraction.
- After assignment, all signalized intersections within the study radius are reported in the affected-intersections table. Project-added trips, v/c ratio change, control delay change, LOS change, and 95th-percentile queue are reported for each intersection so the reviewer can assess relative impact. Intersections beyond the study radius are excluded from analysis.
- Auto-mode share is applied per metro before assignment. Suburban-US metros default to 90% auto (ACS 5-Year B08301 median); transit-heavy metros use measured auto-mode share (e.g., NYC 32%, Tokyo 30%, London 38%, San Francisco 47%). Non-auto trips (transit, walking, cycling) do not load the off-site roadway. This is a screening-level adjustment; a real TIS submittal in a transit-heavy market should refine with project-specific TAZ data.
- Candidate signals are de-duplicated within a 45m clustering threshold to prevent OSM divided-arterial splits and way-record artifacts from double-counting a single physical intersection.
- Intersection-level control delay uses the standard signalized-intersection model $d = d_1 + d_2$ (Webster uniform delay + Akçelik time-dependent incremental-delay term, both openly published) with each intersection's own cycle length and green splits from the signal-timing resolver — a client Synchro record's measured cycle and per-phase splits where the record carries them, otherwise a Webster optimum cycle (60–120 s) with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006), a protected-left phase where the FHWA-HRT-04-091 cross product of left-turn and opposing through volume warrants one, and a pedestrian minimum green from the crossing width; timing is resolved once from no-build volumes and held

fixed across every scenario, approach capacity is $1,800 \text{ vphpl saturation flow} \times \text{that phase's } g/C (\times \text{weather factor})$, and an intersection at or beyond saturation ($Y \geq 0.85$) reports the flat 90s / g/C 0.45 screening default instead, 15-minute peak analysis period ($T = 0.25 \text{ hr}$) and pretimed-signal incremental-delay factor $k = 0.5$. Reported control delay is capped at 300 s (LOS F) for screening reliability — the incremental-delay term is not calibrated far above capacity, so oversaturated approaches are reported as LOS F rather than an implausible delay figure; a calibrated design-level analysis (HCS / Synchro) supersedes this screen.

- Approach-level analysis splits each signal's inflow across NB/SB/EB/WB approaches (each leg from its own volume: the signal's design hour on the two main-road legs — its counted volume where the analyzer joined a compatible count, else the road-class baseline it assigned; each worksheet says which — half per direction, and half in the physical direction of a one-way carriageway; the road-class baseline on uncounted minor legs; a client link count where supplied — each worksheet states the source per leg. Turning movements are then balanced to the exit legs by iterative proportional fitting from a geometry seed, the refinement method of NCHRP Report 255 / NCHRP Report 765, so entries equal exits at every resolved junction; a junction that does not resolve to the road network keeps the screening allocation and says so) and assigns added trips to each approach by cosine-similarity to the bearing of the project relative to the signal. Per-approach v/c, control delay, LOS, and 95th-percentile back-of-queue length (standard cyclic-queue relation with Poisson percentile factor: $Q_{95} \approx Q_1 \times 1.65 \times 25 \text{ ft/veh}$) are reported.

- Level of Service is assigned from the conventional US signalized control-delay bands republished in state DOT traffic-engineering manuals: A $\leq 10s$, B $\leq 20s$, C $\leq 35s$, D $\leq 55s$, E $\leq 80s$, F $> 80s$.

- Sensitivity is reported in narrative form per standard TIA practice: trip-generation method (rate vs. fitted-curve equation), discrete internal capture and pass-by credit variants, and a $\pm 0.5\%/yr$ growth-rate band around the applied value. The engine retains an internal stochastic-sensitivity routine (Box-Muller-perturbed trip rate and existing volume) for demo-mode diagnostics; it is not surfaced in the deliverable because TIA sensitivity is conducted through discrete scoping-agreed scenario variants, not statistical perturbation of unmeasured distributions.

- Mitigations are screening-level recommendations sized to the projected delay change, not full Synchro/SimTraffic optimization runs. A formal TIS submittal should validate these recommendations with detailed traffic counts and signal-timing analysis.

- Turbo-lane (continuous-green-T) screening flags signalized 3-leg T-intersections where one or more main-street through lanes could flow continuously while the minor-street left turn merges in the median — recovering the green time the through movement would otherwise lose to the signal. Candidacy requires measured 3-leg geometry, an arterial-class main street, and a detected median (divided carriageway), all derived from the OpenStreetMap road network. Approach-capacity gain is computed as $(\text{turbo lanes} / \text{approach lanes}) \times (1 - g/C) / (g/C)$, with the main-street through g/C derived from the modeled main-vs-minor critical-flow split; the result is reported within the +7%...+173% envelope documented in the continuous-green-T design literature — the standard national references for this geometry (David Plummer & Associates, Adding Turbo Lanes to T-Intersections, 2010; and the Design Guidelines for the Development of Continuous Green Intersections, 1997). Lane counts and signal timing must be field-verified before design.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is a gamma function $F = a \cdot t^b \cdot e^{-(c \cdot t)}$ — the form recommended by NCHRP Report 716 — on the travel time t to the signal, with home-based-work shape parameters $b = -0.02$, $c = -0.123$ (consistent with NCHRP 716's sampled MPO gravity parameters; the scale constant a cancels in the share normalization).

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Near Cypresswood Drive	0.02	0.1	360	24.8%
Cypresswood Drive & Stuebner Airline Road	0.40	1.0	191	13.2%
Louetta Road & Stuebner Airline Road	0.91	2.2	159	11.0%
Cypresswood Drive & Squyres Road	0.25	0.6	148	10.2%
Squyres Road & Louetta Road	0.87	2.1	108	7.5%
Stuebner Airline Road & Wimbledon Estates Drive	0.58	1.4	103	7.1%
Lyons School Road & Stuebner Airline Road	0.60	1.4	103	7.1%
Louetta Road & Kleinwood Drive	0.85	2.0	52	3.6%
Louetta Road & Lyons School Road	0.87	2.1	52	3.6%
Louetta Road & Theiss Mail Route	0.89	2.1	52	3.6%
Stuebner Airline Road & Strack Road	0.75	1.8	49	3.4%
Cypresswood Drive & Champion Forest Drive	0.84	2.0	46	3.2%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 84% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(\text{ASC} - \lambda \cdot \Delta \text{GC})})$ calibrated to the metro's auto-mode share — measured (ACS B08301) where the metro is wired, otherwise the ACS B08301 metropolitan median of 90% — and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 16% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$, Method of Successive Averages), so congested links shift trips toward alternative routes. 100% of study signals lie on a modeled route; highest loaded-link v/c : 0.33. Corridors carrying the project trips:

Corridor (road class)	Loaded length mi	PM project vph	v/c
Minor arterial	2.79	399	0.33
Collector	0.89	203	0.26
Principal arterial	1.50	149	0.23

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c, delay, LOS, and queue are in the capacity appendix.

Conserved assignment: project trips routed to 21 cordon gateways on the study boundary (weighted by the directional distribution above). Flow conservation verified at render input: Σ entering = Σ leaving at 337 network junctions, max imbalance 0.00 veh. 12 study intersection(s) carry path-derived movements; 2 retain the geometric octant model (labeled per intersection).

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach capacity table. Analysis uses the openly-published Webster/Akcelik signalized control-delay method; v/c , control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Study scope: 14 signalized intersections lie within the 1-mile study area, and all 14 are carried as study intersections. The regional signal inventory holds 36 records inside that radius; 22 were merged into a junction already counted, because a signalized junction is commonly recorded once per approach rather than once per intersection. Merged records are duplicates of an analyzed intersection, not intersections omitted from the study.

Background approach volumes are resolved per leg — the signal's design hour on the main road (its counted volume where the analyzer had a compatible count, else the road-class baseline it assigned — each worksheet says which), the road-class baseline on uncounted legs, client link counts where supplied; half per direction, and half in the physical direction of a one-way carriageway — and the background turning movements in the diagrams are balanced to the exit legs by iterative proportional fitting (NCHRP 255/765 refinement) from a geometry seed; each worksheet states its leg sources and residual. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Near Cypresswood Drive

Signal ID

Location

Added PM peak trips

Existing / No-Build (PM)

Build (PM)

Δ control delay

Mitigation

houston-1743

NW Houston-The Woodlands-Sugar Land · 0.02 mi from site

360

LOS C · 34.4 s/veh · v/c 0.94

LOS F · 89.5 s/veh · v/c 1.13

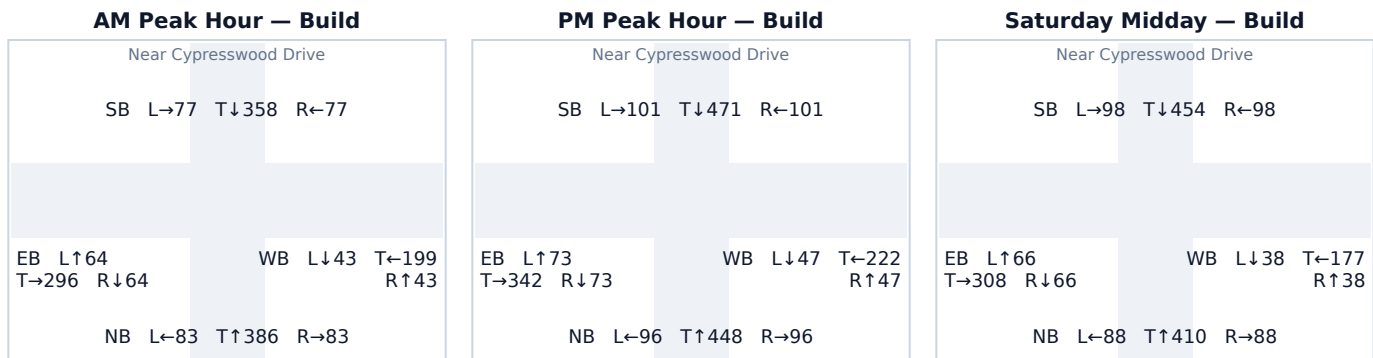
+55.1 s/veh (LOS change)

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +55.1s, design year +83.6s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.47 / EW 0.36; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: screening allocation (this junction did not resolve to the road network); turning movements: screening 15/70/15.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	537	104	640	0.64 → 0.76	15.8 → 19.5	B → B	363
SB	486	187	673	0.58 → 0.80	14.5 → 21.4	B → C	393
EB	419	69	488	0.64 → 0.74	20.4 → 24.0	C → C	292
WB	316	0	316	0.48 → 0.48	17.2 → 17.2	B → B	168

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	112	31%
NB Through	104	29%
SB Right	75	21%
EB Left	69	19%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

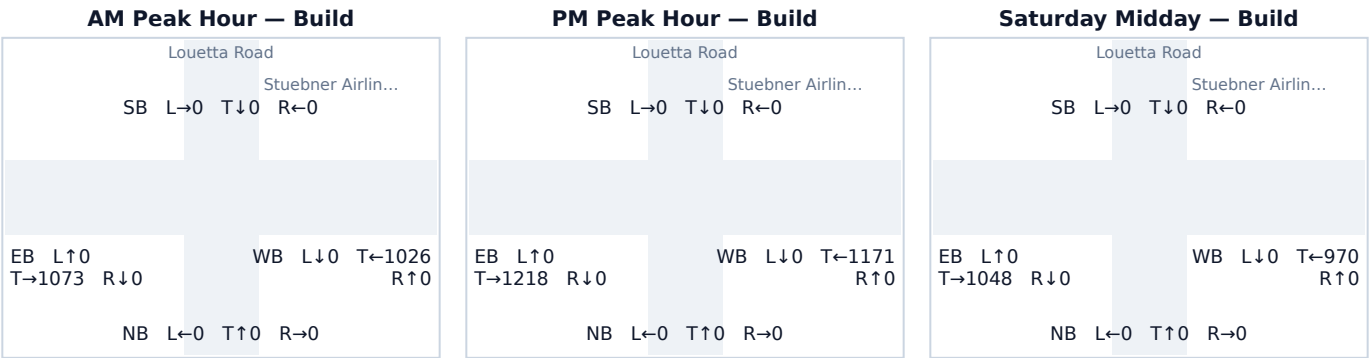
A.2 Louetta Road & Stuebner Airline Road

Signal ID	houston-723
Location	NW Houston-The Woodlands-Sugar Land · 0.91 mi from site
Added PM peak trips	159
Existing / No-Build (PM)	LOS C · 25.1 s/veh · v/c 0.93
Build (PM)	LOS D · 37.1 s/veh · v/c 0.99
Δ control delay	+12.0 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +12.0s, design year +28.5s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.23 / EW 0.60; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 2 from the signal's counted design hour (half per direction). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph). 2 extra leg not carried (4×4 matrix).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ EB leg	→ WB leg	Σ entering
EB approach	—	1,099	1,099
WB approach	1,099	—	1,099
Σ exiting	1,099	1,099	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
SB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
EB	1,115	103	1,218	1.03 → 1.12	46.6 → 79.7	D → E	825
WB	1,115	56	1,171	1.03 → 1.08	46.6 → 63.3	D → E	793

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	103	65%
WB Left	37	23%
WB Through	19	12%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

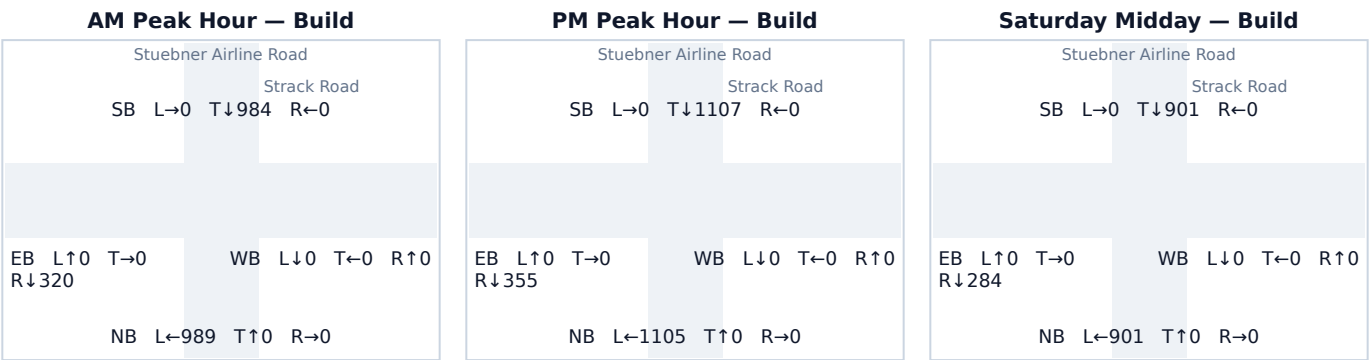
A.3 Stuebner Airline Road & Strack Road

Signal ID	houston-696
Location	NW Houston-The Woodlands-Sugar Land · 0.75 mi from site
Added PM peak trips	49
Existing / No-Build (PM)	LOS F · 211.8 s/veh · v/c 1.40
Build (PM)	LOS F · 223.7 s/veh · v/c 1.43
Δ control delay	+11.9 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +11.8s, design year +0.0s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 1.40 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: did not balance within 50 iterations (residual 452.5 vph; rows held exact); exits scaled to entries, 43% imbalance. 1 extra leg not carried (4×4 matrix). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	Σ entering
NB approach	—	0	1,066	1,066
SB approach	1,066	—	0	1,066
EB approach	350	0	—	350
Σ exiting	1,416	0	1,066	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	1,082	24	1,105	1.34 → 1.36	184.0 → 196.5	F → F	1,131
SB	1,082	25	1,107	1.34 → 1.37	184.0 → 197.6	F → F	1,133
EB	355	0	355	0.44 → 0.44	18.7 → 18.7	B → B	251
WB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	25	51%
NB Through	24	49%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

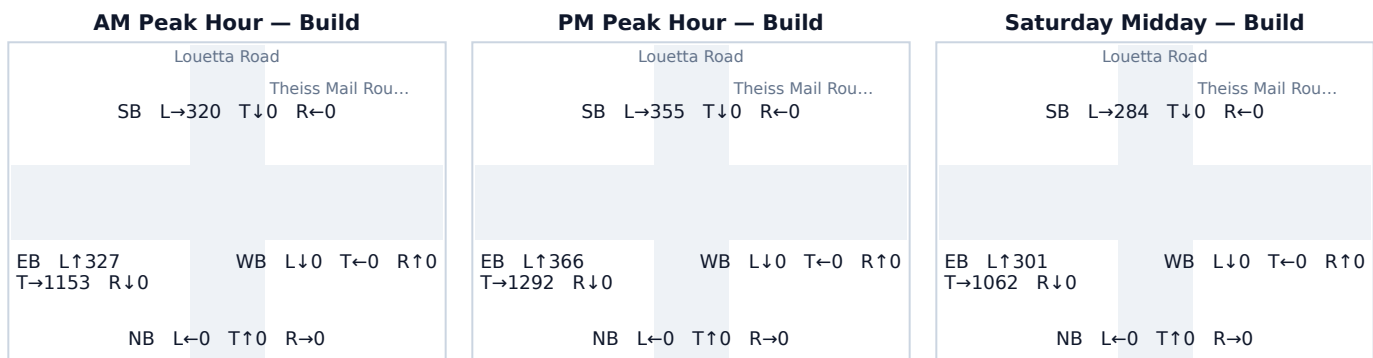
A.4 Louetta Road & Theiss Mail Route

Signal ID	houston-5279
Location	NW Houston-The Woodlands-Sugar Land · 0.89 mi from site
Added PM peak trips	52
Existing / No-Build (PM)	LOS F · 83.3 s/veh · v/c 1.09
Build (PM)	LOS F · 93.9 s/veh · v/c 1.12
Δ control delay	+10.6 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +10.6s, design year +12.7s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 1.09 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.0 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ SB leg	→ EB leg	→ WB leg	Σ entering
SB approach	—	0	350	350
EB approach	350	—	1,232	1,582
WB approach	0	0	—	0
Σ exiting	350	0	1,582	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
SB	355	0	355	0.44 → 0.44	18.7 → 18.7	B → B	251
EB	1,606	52	1,658	1.98 → 2.05	300.0 → 300.0	F → F	1,696
WB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	52	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

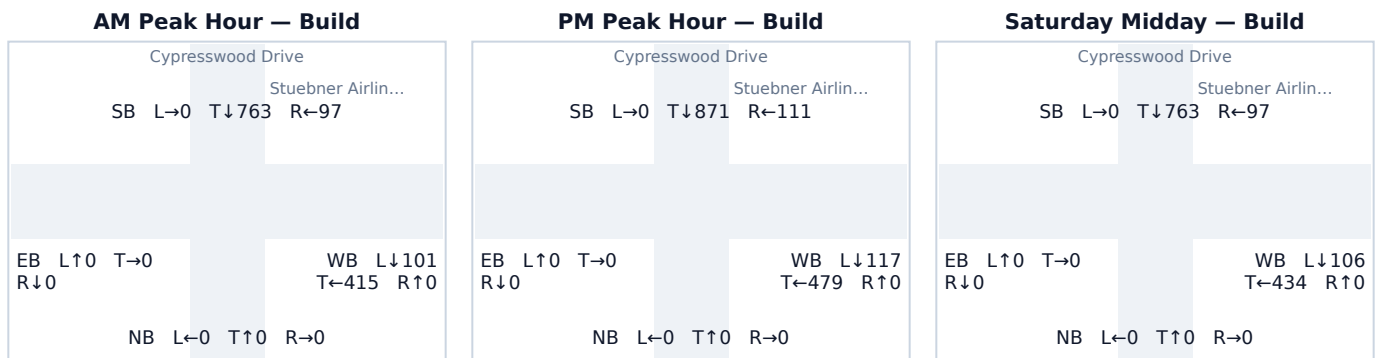
A.5 Cypresswood Drive & Stuebner Airline Road

Signal ID	houston-4193
Location	NW Houston-The Woodlands-Sugar Land · 0.40 mi from site
Added PM peak trips	191
Existing / No-Build (PM)	LOS B · 15.9 s/veh · v/c 0.62
Build (PM)	LOS B · 18.2 s/veh · v/c 0.70
Δ control delay	+2.3 s/veh
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required. Governing horizon: design year (opening year +2.3s, design year +7.6s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 90 s cycle, 2 critical phases; g/C NS 0.56 / EW 0.33; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 7 iterations, residual 0.3 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	0	0	0	0
SB approach	768	—	98	0	866
EB approach	0	0	—	0	0
WB approach	98	0	402	—	500
Σ exiting	866	0	500	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	8.6 → 8.6	A → A	0
SB	879	103	982	0.87 → 0.97	26.6 → 40.5	C → D	973
EB	0	0	0	0.00 → 0.00	20.5 → 20.5	C → C	0
WB	508	88	596	0.87 → 1.02	44.3 → 71.8	D → E	611

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Right	103	54%
WB Through	88	46%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

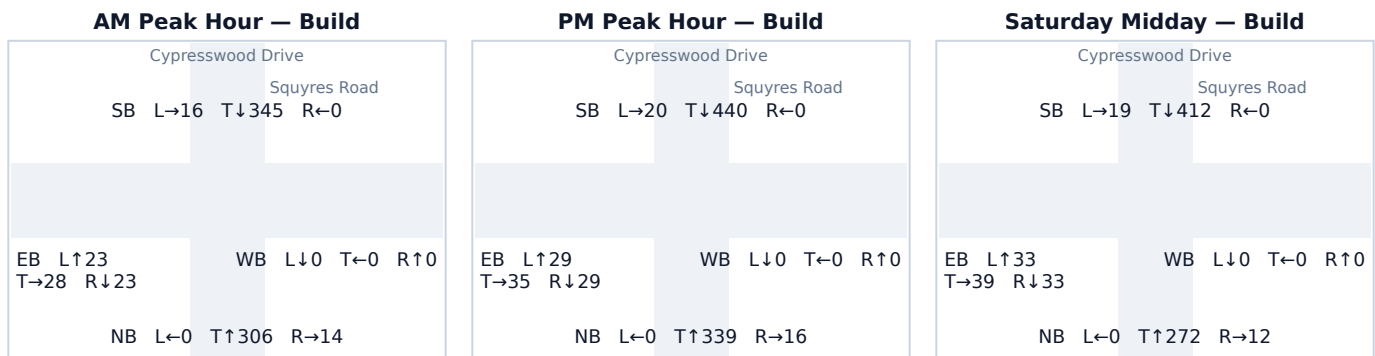
A.6 Cypresswood Drive & Squyres Road

Signal ID	houston-4050
Location	NW Houston-The Woodlands-Sugar Land · 0.25 mi from site
Added PM peak trips	148
Existing / No-Build (PM)	LOS A · 6.6 s/veh · v/c 0.32
Build (PM)	LOS A · 7.1 s/veh · v/c 0.38
Δ control delay	+0.5 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.5s, design year +0.7s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.60 / EW 0.23; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.1 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	335	0	16	350
SB approach	335	—	0	16	350
EB approach	16	16	—	19	50
WB approach	0	0	0	—	0
Σ exiting	350	350	0	50	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	0	355	0.33 → 0.33	6.7 → 6.7	A → A	121
SB	355	105	460	0.33 → 0.42	6.7 → 7.6	A → A	169
EB	51	43	93	0.12 → 0.22	18.9 → 20.0	B → B	52
WB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Left	105	71%
EB Through	43	29%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

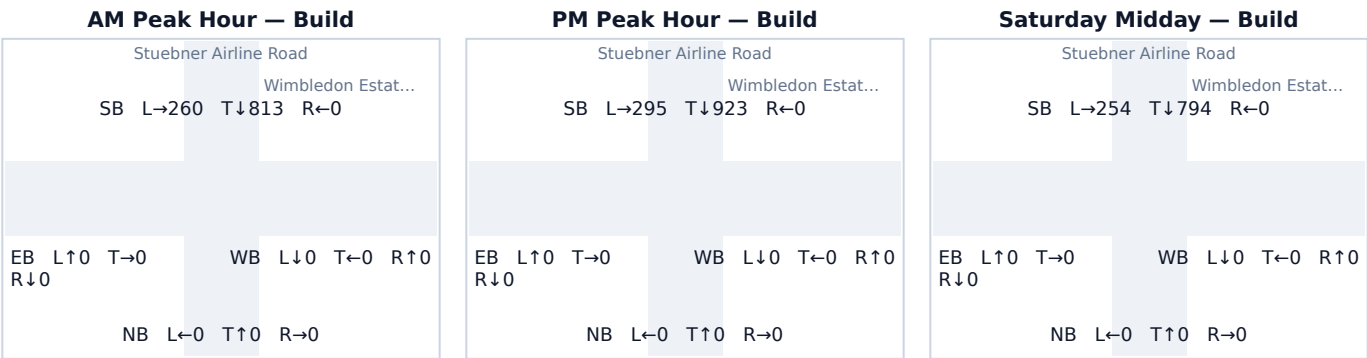
A.7 Stuebner Airline Road & Wimbledon Estates Drive

Signal ID	houston-4757
Location	NW Houston-The Woodlands-Sugar Land · 0.58 mi from site
Added PM peak trips	103
Existing / No-Build (PM)	LOS A · 8.0 s/veh · v/c 0.46
Build (PM)	LOS A · 8.5 s/veh · v/c 0.51
Δ control delay	+0.5 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.5s, design year +0.9s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.60 / EW 0.23; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph; exits scaled to entries, 32% imbalance). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ WB leg	Σ entering
NB approach	—	0	0	0
SB approach	833	—	265	1,099
WB approach	0	0	—	0
Σ exiting	833	0	265	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	4.7 → 4.7	A → A	0
SB	1,115	103	1,218	1.03 → 1.12	46.6 → 79.7	D → E	825
EB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
WB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	103	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 Lyons School Road & Stuebner Airline Road

Signal ID	houston-1746
Location	NW Houston-The Woodlands-Sugar Land · 0.60 mi from site
Added PM peak trips	103
Existing / No-Build (PM)	LOS A · 6.6 s/veh · v/c 0.32
Build (PM)	LOS A · 7.0 s/veh · v/c 0.36
Δ control delay	+0.4 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.4s, design year +0.5s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.23 / EW 0.60; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.1 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	0	0	0	0
SB approach	19	—	16	16	50
EB approach	16	0	—	335	350
WB approach	16	0	335	—	350
Σ exiting	50	0	350	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
SB	51	103	154	0.12 → 0.37	18.9 → 21.9	B → C	89
EB	355	0	355	0.33 → 0.33	6.7 → 6.7	A → A	121
WB	355	0	355	0.33 → 0.33	6.7 → 6.7	A → A	121

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	103	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 Squyres Road & Louetta Road

Signal ID

Location

Added PM peak trips

Existing / No-Build (PM)

Build (PM)

Δ control delay

Mitigation

houston-5274

NW Houston-The Woodlands-Sugar Land · 0.87 mi from site

108

LOS A · 6.9 s/veh · v/c 0.35

LOS A · 7.3 s/veh · v/c 0.39

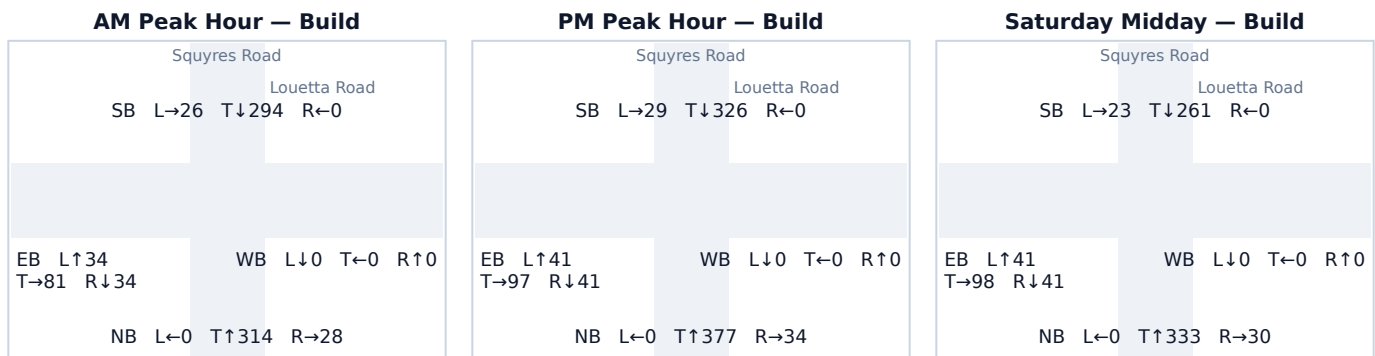
+0.4 s/veh

No mitigation indicated at screening level (opening year +0.4s, design year +0.5s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.60 / EW 0.23; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 4 iterations, residual 0.1 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	321	0	29	350
SB approach	321	—	0	29	350
EB approach	29	29	—	68	125
WB approach	0	0	0	—	0
Σ exiting	350	350	0	125	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	56	411	0.33 → 0.38	6.7 → 7.2	A → A	146
SB	355	0	355	0.33 → 0.33	6.7 → 6.7	A → A	121
EB	127	52	179	0.31 → 0.43	21.0 → 22.9	C → C	105
WB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	56	52%
EB Through	52	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

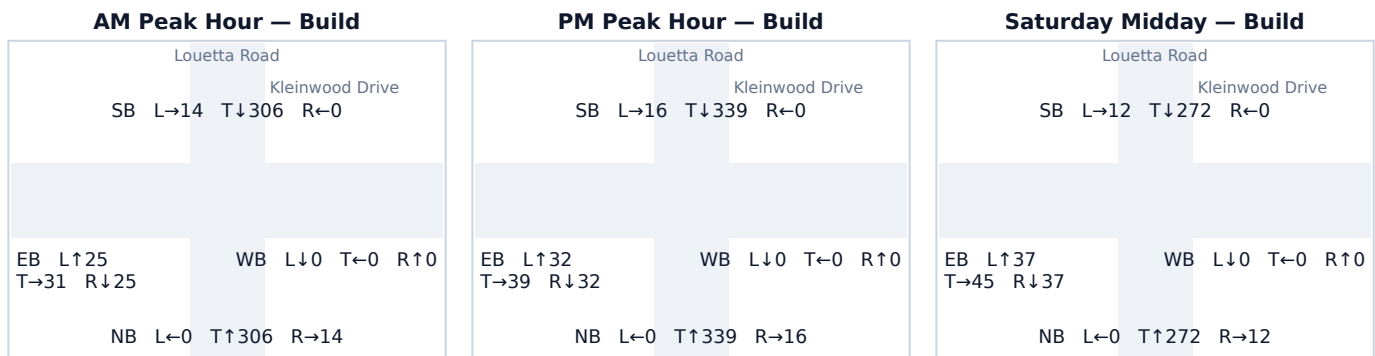
A.10 Louetta Road & Kleinwood Drive

Signal ID	houston-247
Location	NW Houston-The Woodlands-Sugar Land · 0.85 mi from site
Added PM peak trips	52
Existing / No-Build (PM)	LOS A · 6.6 s/veh · v/c 0.32
Build (PM)	LOS A · 6.8 s/veh · v/c 0.34
Δ control delay	+0.2 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.2s, design year +0.2s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.60 / EW 0.23; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.1 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	335	0	16	350
SB approach	335	—	0	16	350
EB approach	16	16	—	19	50
WB approach	0	0	0	—	0
Σ exiting	350	350	0	50	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	0	355	0.33 → 0.33	6.7 → 6.7	A → A	121
SB	355	0	355	0.33 → 0.33	6.7 → 6.7	A → A	121
EB	51	52	103	0.12 → 0.25	18.9 → 20.2	B → C	58
WB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	52	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

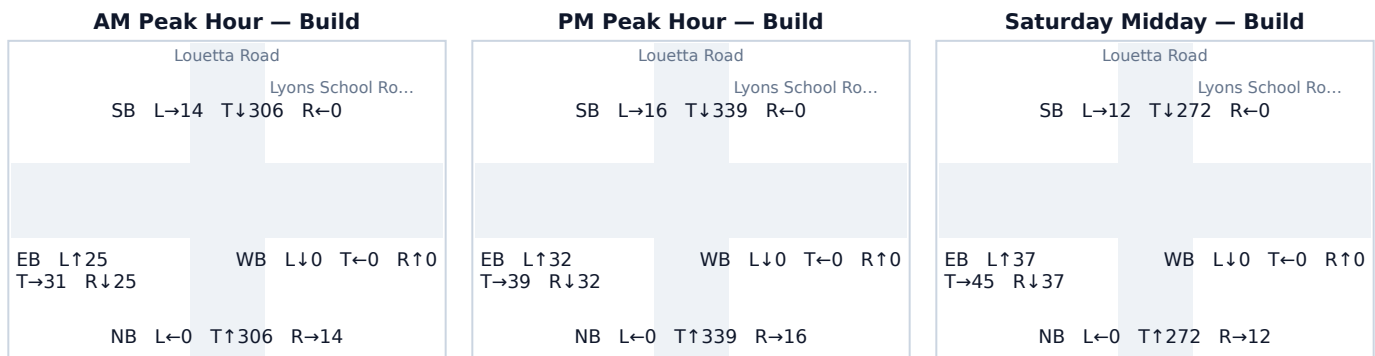
A.11 Louetta Road & Lyons School Road

Signal ID	houston-200
Location	NW Houston-The Woodlands-Sugar Land · 0.87 mi from site
Added PM peak trips	52
Existing / No-Build (PM)	LOS A · 6.6 s/veh · v/c 0.32
Build (PM)	LOS A · 6.8 s/veh · v/c 0.34
Δ control delay	+0.2 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.2s, design year +0.2s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.60 / EW 0.23; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.1 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	335	0	16	350
SB approach	335	—	0	16	350
EB approach	16	16	—	19	50
WB approach	0	0	0	—	0
Σ exiting	350	350	0	50	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	0	355	0.33 → 0.33	6.7 → 6.7	A → A	121
SB	355	0	355	0.33 → 0.33	6.7 → 6.7	A → A	121
EB	51	52	103	0.12 → 0.25	18.9 → 20.2	B → C	58
WB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	52	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

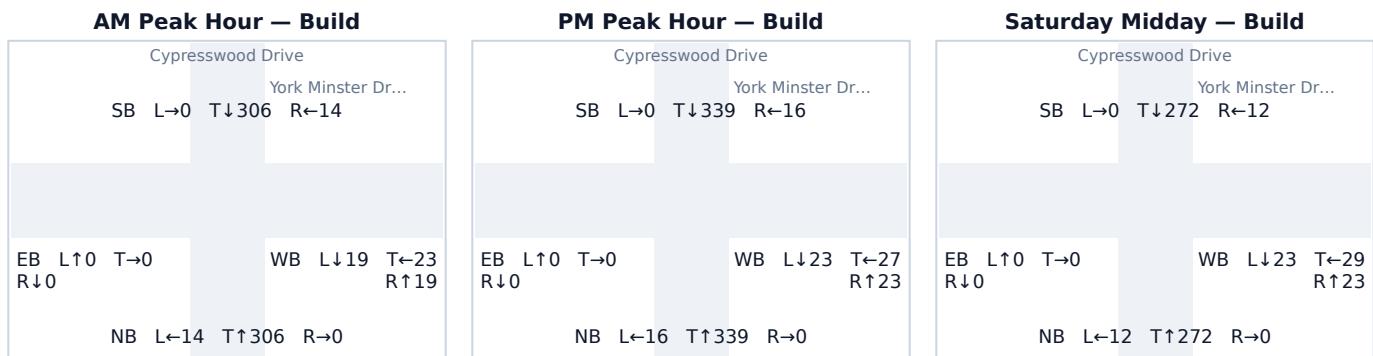
A.12 Cypresswood Drive & York Minster Drive

Signal ID	houston-2008
Location	NW Houston-The Woodlands-Sugar Land · 0.65 mi from site
Added PM peak trips	22
Existing / No-Build (PM)	LOS A · 6.6 s/veh · v/c 0.32
Build (PM)	LOS A · 6.7 s/veh · v/c 0.33
Δ control delay	+0.1 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.1s, design year +0.1s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.60 / EW 0.23; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.1 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	335	16	0	350
SB approach	335	—	16	0	350
EB approach	0	0	—	0	0
WB approach	16	16	19	—	50
Σ exiting	350	350	50	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	0	355	0.33 → 0.33	6.7 → 6.7	A → A	121
SB	355	0	355	0.33 → 0.33	6.7 → 6.7	A → A	121
EB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
WB	51	22	73	0.12 → 0.18	18.9 → 19.4	B → B	40

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	22	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

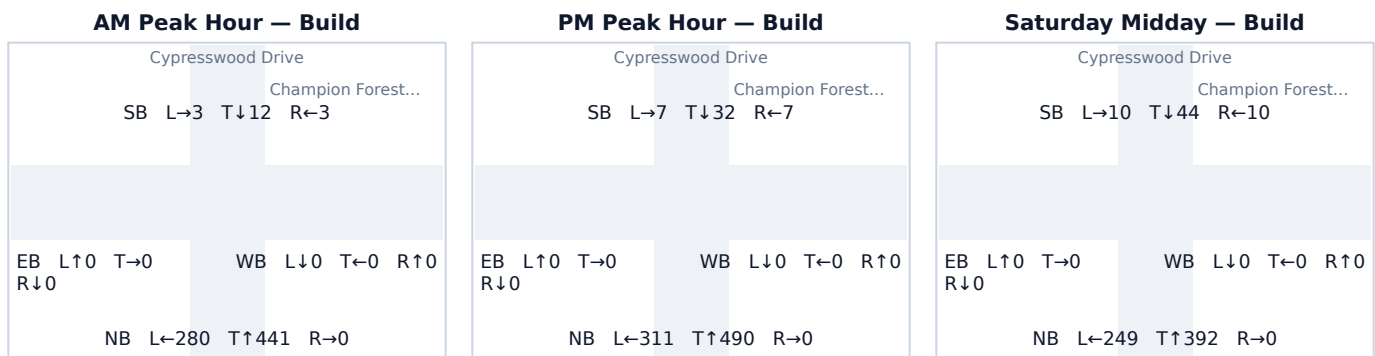
A.13 Cypresswood Drive & Champion Forest Drive

Signal ID	houston-445
Location	NW Houston-The Woodlands-Sugar Land · 0.84 mi from site
Added PM peak trips	46
Existing / No-Build (PM)	LOS A · 6.8 s/veh · v/c 0.33
Build (PM)	LOS A · 6.9 s/veh · v/c 0.35
Δ control delay	+0.1 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.2s, design year +0.2s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.60 / EW 0.23; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph; exits scaled to entries, 63% imbalance). 1 extra leg not carried (4×4 matrix). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	Σ entering
NB approach	—	483	306	790
SB approach	0	—	0	0
EB approach	0	0	—	0
Σ exiting	0	483	306	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	801	0	801	0.74 → 0.74	13.1 → 13.1	B → B	395
SB	0	46	46	0.00 → 0.04	4.7 → 4.9	A → A	13
EB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
WB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Right	46	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

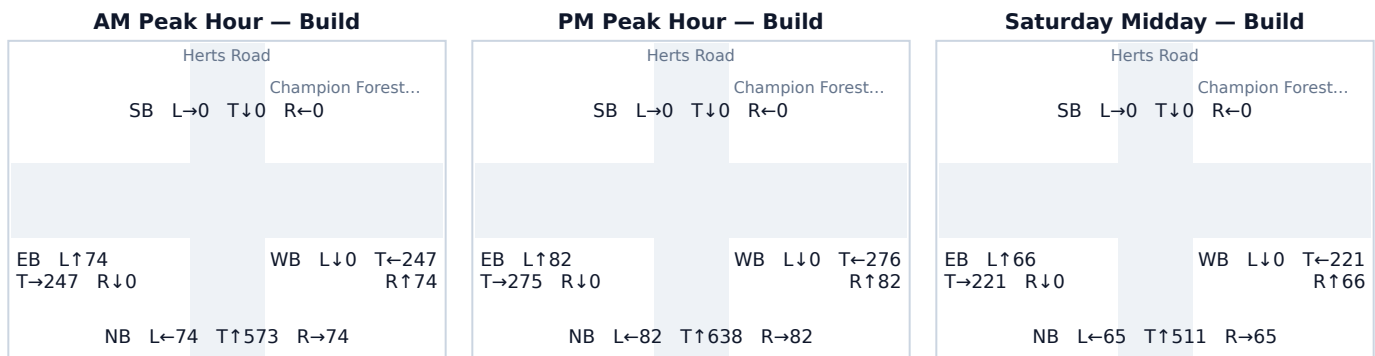
A.14 Herts Road & Champion Forest Drive

Signal ID	houston-446
Location	NW Houston-The Woodlands-Sugar Land · 0.90 mi from site
Added PM peak trips	4
Existing / No-Build (PM)	LOS B · 11.8 s/veh · v/c 0.65
Build (PM)	LOS B · 11.9 s/veh · v/c 0.66
Δ control delay	+0.1 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.2s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.58 / EW 0.26; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 6 iterations, residual 0.1 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	629	80	80	790
SB approach	0	—	0	0	0
EB approach	0	80	—	270	350
WB approach	0	80	270	—	350
Σ exiting	0	789	350	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	801	0	802	0.77 → 0.77	15.2 → 15.2	B → B	420
SB	0	0	0	0.00 → 0.00	5.4 → 5.4	A → A	0
EB	355	2	357	0.77 → 0.77	32.5 → 32.7	C → C	228
WB	355	2	358	0.77 → 0.78	32.5 → 32.8	C → C	228

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	2	50%
EB Through	2	50%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.